IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 4 November 2025, accepted 21 November 2025, date of publication 1 December 2025,
date of current version 15 December 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3639087

RESEARCH ARTICLE

CQELedu: Design and Implementation of a LangChain and GPT-4o Mini-Based Web Application for Custom Question Generation and Error-Based Learning in Education

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This work was supported by the National Research Foundation of Korea (NRF) grant, funded by the Korea government (MSIT) (No. RS-2024-00455788).

ABSTRACT Evaluation and feedback are central to the learning process, making it essential to analyze learners' incorrect answers and generate tailored questions or supplementary materials that address their weaknesses and improve learning outcomes. This paper presents CQELedu, a web application designed to generate questions based on learning materials and analyze incorrect responses to create error-driven follow-up questions. The proposed system leverages GPT-4o mini and LangChain to generate both subjective and objective questions from user-provided materials. The system analyzes incorrect answers and generates customized follow-up questions, helping learners identify and address gaps in understanding. The relevance, accuracy, and response time of the generated questions are assessed both quantitatively and qualitatively to evaluate system performance. Experimental results demonstrate high cosine similarity scores for relevance, validating the relevance of the generated questions to the learning materials. The accuracy of the system is reflected in the applicability of the questions derived from incorrect answers. Additionally, the system exhibits improved response time compared to existing methods, highlighting its efficiency. Thus, CQELedu is well-suited for supporting dynamic learning environments, particularly in standardized testing contexts such as certification exams and school assessments. Future research should explore the integration of learning materials from diverse fields, the implementation of multilingual support, and the scalability potential, aiming to expand the applicability of CQELedu across various educational domains.

INDEX TERMS Artificial intelligence, educational technology, GPT, large language models, personalized learning, question generation.

I. INTRODUCTION

Advancements in artificial intelligence (AI) technology have produced transformative changes across the educational ecosystem, including in the areas of content creation, question generation, automated grading, and feedback delivery [1], [2], [3], [4]. Education systems have evolved in their core functionalities through the incorporation of AI technology, moving beyond the delivery of educational content to encompass analyzing learning data, generating questions, and offering customized feedback, establishing a precedent for education driven by AI technology. The introduction of intelligent tutoring systems, automated assessment tools, content generation and management, and learner behavior analysis technologies have served to enhance the quality of the educational experience and improve learning efficiency [5], [6], [7], [8], [9]. Recent studies have focused on developing and evaluating the effectiveness of AI-based

The associate editor coordinating the review of this manuscript and approving it for publication was Maria Chiara Caschera.

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personalized learning systems, with the primary objective of delivering individualized learning experiences that align with learners' unique requirements and capabilities [10], [11].

Large Language Models (LLMs), which play a pivotal role in analyzing key information from learning materials and generating contextually appropriate questions [12], [13], have been central to these developments. Using extensive training data, LLMs address learners' needs by providing customized questions and answers, thereby creating a dynamic and adaptive learning experience [14], [15]. These models extract meaningful insights from uploaded source material and generate questions aligned with learners' goals and proficiency levels [16]. By moving beyond traditional static question banks, LLM-based systems offer flexible and interactive learning experiences that effectively cater to learners' needs [17], [18].

LangChain enhances the functionality of LLMs by streamlining workflows, optimizing efficiency, and improving the quality of the data processing and question generation processes [19], [20]. Integrating LangChain with LLMs simplifies the generation of complex questions, ensures alignment with learners' needs, and fosters a personalized learning environment. Consequently, learners benefit from structured and adaptive learning experiences based on their performance and data.

Research on the educational applications of LLMs and LangChain has explored diverse functionalities, such as summarizing core information from topic-specific PDFs and generating questions relevant to uploaded source matter [21], [22]. Additionally, other studies have focused on analyzing learner response data to provide real-time, personalized chatbot support [16], [23], [24]. Evaluation and feedback are critical elements in the learning process as they enhance comprehension and promote deep learning.

Analyzing learners' incorrect answers is particularly vital for improving learning outcomes. Error-based learning (EBL) is an instructional approach that aims to maximize learning effectiveness by encouraging learners to actively identify, analyze, and correct errors encountered during the learning process [25]. Error analysis provides learners with new learning opportunities and helps them to recognize and understand their weaknesses and misconceptions [26]. By generating tailored questions or supplementary materials based on incorrect answers, learners can systematically address their weaknesses [27], [28].

Although this feedback method has proven to be effective, there is limited literature on the subject, with existing research focusing primarily on question generation and system implementation [17], [29], [30]. Systematic evaluations of generated question quality are scarce, especially those verifying alignment with source materials and assessing educational value using quantitative and qualitative metrics [31], [32].

To address these deficits, this study introduces CQELedu, a web application designed to generate questions based on learning materials and analyze learners' responses to create error-driven questions that address their weaknesses. The proposed system utilizes GPT-4o mini and LangChain to generate both subjective and objective questions from user-uploaded materials.

Further, the system provides tailored error-driven questions by analyzing incorrect answers submitted by learners. System effectiveness is validated through quantitative and qualitative evaluations, focusing on question accuracy, relevance, and response time. Beyond simple question generation, CQELedu supports learners by providing customized questions and supplementary materials based on their specific challenges. By identifying specific topics and addressing areas of difficulty faced by learners, CQELedu enables learners to systematically improve their understanding and performance through the generation of supplementary materials and additional questions.

This study not only enhances personalized learning experiences but also establishes a new direction for advancing educational technology. By addressing learners' weaknesses and maximizing learning outcomes, CQELedu transcends traditional static learning methods, creating a dynamic and adaptive learning environment that contributes to the future of educational technologies.

II. RELATED WORK

A. RESEARCH ON LLM-BASED EDUCATIONAL SYSTEMS

The advanced language comprehension and generative capabilities of LLMs have catalyzed active research into their applications in education. Studies have demonstrated the potential of LLMs in various educational domains, including automatic question generation, feedback provision, and assessment processes.

Kirova et al. [33] analyzed the impact of LLMs on software engineering education and emphasized the necessity of automating question generation and evaluation by integrating them into the curriculum. Meißner et al. [34] proposed EvalQuiz, a system to automatically generate self-assessment quizzes based on lecture materials using GPT-4. While the system simplified quiz generation and reduced time requirements, it exhibited limitations in question originality, diversity, and occasional logical inconsistencies.

Hang et al. [18] introduced and evaluated MCQGen, a system combining GPT-4 with retrieval-augmented generation (RAG) to generate multiple-choice questions (MCQs) tailored to learners' needs. Despite its support for personalized learning and improved academic achievement, the system faced challenges in domain-specific applications and multilingual support. Olney [35] assessed the performance of LLMs in generating MCQs from textbooks and found similarities with human-generated questions. However, issues such as duplicate answer choices and missing correct answers indicated the need for error reduction.

Recent studies have explored adaptive feedback using LLMs, leveraging their advanced language capabilities.

TABLE 1. Comparison of pricing and token capacities between GPT-4o and GPT-4o mini.

Model	Pricing	Context window	Max output tokens
GPT-4o	\$2.50 / 1M input tokens \$10.00 / 1M output tokens	128,000 tokens	16,384 tokens
GPT-4o mini	\$0.150 / 1M input tokens \$0.600 / 1M output tokens	128,000 tokens	16,384 tokens

These approaches go beyond question generation to provide personalized feedback to learners.

Kabir and Lin proposed a personalized adaptive learning system that utilizes ChatGPT to automatically generate questions and feedback tailored to the learner's knowledge level, and to assess their mastery [36].

Andonov proposed LLM-based personalized feedback for programming assignments and demonstrated the feasibility of generating meaningful feedback [37]. Agostini and Picasso proposed an educational system that utilizes LLMs to enable sustainable and authentic assessment and feedback in higher education [38].

Collectively, existing studies have demonstrated the feasibility of LLM-based automatic question-generation systems in education while emphasizing the need for improvements, including domain expansion and error resolution.

B. RESEARCH ON LLM-BASED EDUCATIONAL AND PDF INTERACTION SYSTEMS USING LANGCHAIN

LangChain, an open-source framework, facilitates the development of LLM-based applications and is widely applied in the education field and in chatbot development.

The framework simplifies the creation of complex workflows by offering modular components, such as document loaders, embeddings, vector stores, and prompt templates. Integrating RAG, LangChain enables LLMs to interact with external knowledge bases, thereby enhancing domain-specific accuracy [39], [40], [41]. Additionally, the memory capabilities of LLMs promote long-term conversational context, supporting the development of personalized learning tools and maximizing system scalability and efficiency.

Numerous studies have demonstrated the potential of LangChain-based systems for interacting with PDFs [16], [21], [22], [23], [24]. Sar et al. [23] developed and evaluated a PDF-based interactive chatbot using LLAMA2 and LangChain to achieve high accuracy and efficiency in document-based question answering. However, the system required improvements in handling large and complex documents. Khadija et al. [32] designed a PDF-based automatic information retrieval chatbot using GPT-3.5 Turbo. Although the system accurately answered user queries, it required enhancements in complex data processing. Neira-Maldonado et al. [29] utilized GPT-3.5 Turbo and LangChain to develop an intelligent educational agent (IEA) that generated questions, provided personalized learning support, and offered automated assessments. Despite its capability to deliver personalized learning experiences and its demonstrated efficacy as an educational tool, the system required better reasoning explanations and improved computational infrastructure. Sheremet et al. [42] developed a PDF-based interactive system using GPT-4 to extract meaningful information and generate responses. While the system improved data analysis productivity, the study emphasized the importance of building user-friendly interfaces. Therefore, existing research has underscored the accuracy and efficiency of LangChain-based LLM web applications in education, while emphasizing areas for improvement, such as data scalability, interpretability, and user-friendly interface development.

These advancements are crucial for optimizing automatic question generation, feedback provision, and assessment processes in educational settings.

III. PROPOSED METHOD

A. CQELedu: WEB-BASED LEARNING SUPPORT PLATFORM FOR CUSTOM QUESTION GENERATION USING LLM TECHNOLOGY

CQELedu is a web-based learning support platform that leverages LangChain and LLMs to create customized questions based on user-uploaded files. The system aims to deliver document and error-driven questions and real-time feedback during the learning process. Accordingly, the platform employs GPT-4o mini, a lightweight OpenAI LLM, to enhance operational efficiency by reducing response time, resource use, and token costs.

Table 1 compares OpenAI's GPT-4o and GPT-4o mini models in terms of token pricing, context window size, and maximum output token limits [43], [44]. GPT-4o mini offers a relatively low token cost, enabling cost-effective operation in environments requiring large-scale question generation and real-time interaction. The model supports both text and image inputs while generating structured text outputs. It performs well on text and multimodal reasoning at lower cost and latency and can refine outputs from larger models such as GPT-4o. In CQELedu, GPT-4o mini serves as the LLM model for generating questions and facilitating real-time chatbot interactions, efficiently handling document and error-based questions, along with real-time chatbot feedback.

LangChain, an open-source framework designed to simplify the development of LLM-based applications, plays a central role in the implementation of CQELedu. It supports various tools including prompt templates, embeddings, and output parsers, enhancing efficiency and streamlining

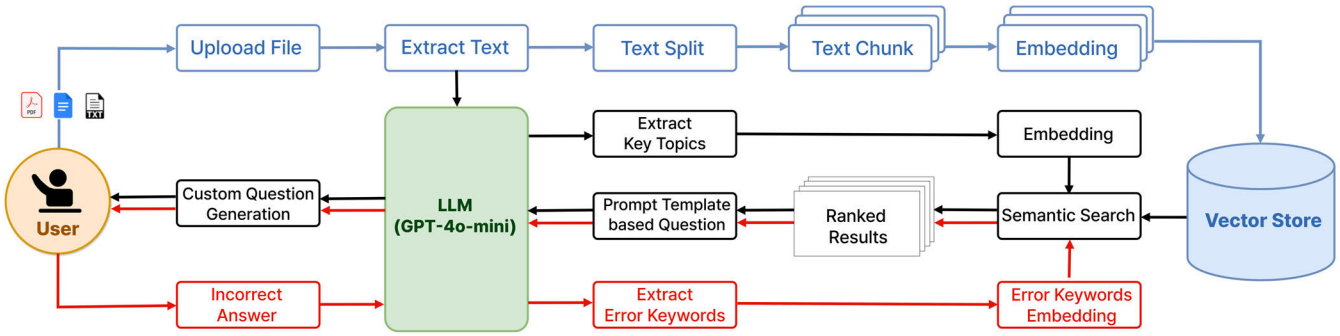


FIGURE 1. CQELedu system architecture.

development for tasks such as chatbot creation and document analysis. CQELedu uses LangChain to design its overall system structure and data flow. The system architecture and data flow are illustrated in Fig. 1. The key components of the system are discussed as follows:

1) DOCUMENT UPLOAD AND TEXT PREPROCESSING

Users upload learning materials in supported formats, such as PDF, DOCX, or TXT. To protect user privacy, the system only receives and processes the minimum amount of data necessary for question generation. The uploaded files are stored in a designated data directory and processed into text using PyPDF2 and docx2txt libraries. Considering the token limit of LLMs, text is segmented into chunks using LangChain's CharacterTextSplitter, which maintains context and minimizes information loss. These chunks are used for similarity searches and question generation.

2) EMBEDDING AND VECTOR STORE STORAGE

The segmented chunks are embedded into vectors using OpenAI's text-embedding-ada-002 model, which quantifies the semantic similarity between chunks. The resulting vectors are stored in a chroma vector store, enabling efficient and accurate retrieval of chunks related to user queries.

3) QUESTION GENERATION

When a document is uploaded, GPT-4o mini extracts core concepts from the content. These core concepts are embedded using text-embedding-ada-002 model and compared with document chunks using cosine similarity. The top-ranked chunks based on similarity are returned as ranked results. This process is implemented using a Retrieval-Augmented Generation (RAG) approach. It retrieves relevant content from user-uploaded documents to ground the model, reducing reliance on the LLM's pre-trained knowledge. This design structurally mitigates the risk of hallucinations and bias, particularly in educational contexts. Further, users can select the question type (multiple-choice or subjective), based on which LangChain's PromptTemplate generates custom questions. The extracted core concepts and top-ranked chunks ensure that the generated questions align with the context of the uploaded documents. We apply FewShotPromptTemplate to enforce output format and patterns, improving consistency. This also mitigates prompt sensitivity and yields more stable outputs.

4) ANSWER GENERATION

The generated PromptTemplate is fed into the GPT-4o mini, which produces the final answer. OpenAI automatically detects harmful content, including discrimination, violence, or illegal activities related to factors such as race or gender. When such content is detected, the system immediately halts question generation to prevent the production of inappropriate feedback. To further reduce the likelihood of harmful outputs, safety-oriented prompts were integrated into the system to minimize the risk of generating inappropriate content. The output includes questions, correct answers, distractors, and reference materials. The responses are parsed and organized using LangChain's Output Parser to ensure clear separation and structure. Reference materials aid in understanding the context of questions, while distractors are utilized in subsequent processes to generate error-driven questions.

The proposed design allows users to verify correct answers, understand incorrect ones, and focus on reinforcing weak concepts.

5) ERROR-BASED QUESTION GENERATION

The Output Parser deconstructs question-solving results, enabling the creation of error-based questions. Incorrect answers are analyzed to identify misunderstood concepts. The incorrect answer keywords are embedded using the text-embedding-ada-002 model and compared with document embeddings in the Chroma vector store through cosine similarity. The most relevant document chunks are ranked and returned as the ranked results. Based on the ranked chunks, LangChain's PromptTemplate generates error-based questions. FewShotPromptTemplate ensures that the new questions maintain consistency in format and pattern with the previously generated ones. This process enables users to engage with new questions based on earlier mistakes, thus facilitating personalized learning. Unlike the simple

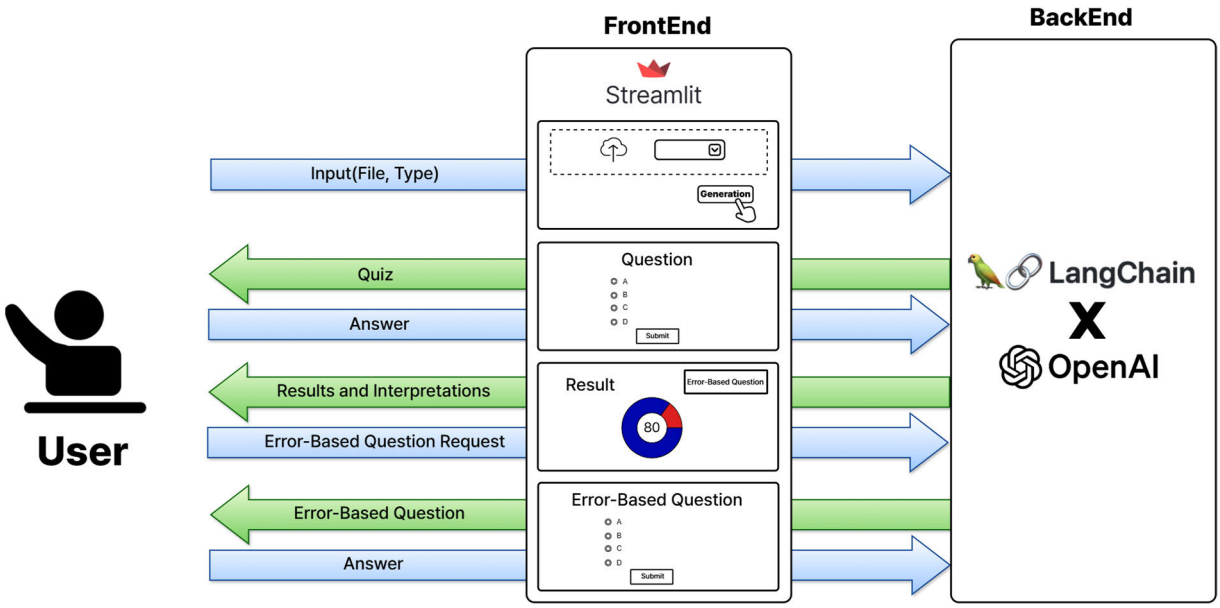


FIGURE 2. CQELedu interaction process.

repetition of previous questions, the system generates novel questions based on incorrect answers, enabling targeted reinforcement of weak concepts.

B. IMPLEMENTATION OF WEB APPLICATION

CQELedu is implemented as a file-based question-generation system using Streamlit, a Python framework that enables efficient development of user-friendly web applications. Streamlit provides an intuitive interface that enables seamless integration between LangChain, LLM tasks, and the backend, thereby enhancing user interaction and experience.

The primary features of the CQELedu web application are discussed as follows:

1) FILE UPLOAD AND PREPROCESSING

Users upload learning material files, which are converted into text and segmented into chunks for processing.

2) CUSTOM QUESTION GENERATION

Based on the selected question type (multiple-choice or subjective), the PromptTemplate is used to generate tailored questions.

3) ERROR-BASED QUESTION GENERATION

After answering the questions, the system generates error-based questions to support iterative learning.

4) CHATBOT INTERACTION

During the learning process, users can interact with chatbots to ask questions and receive real-time responses.

The architecture of the CQELedu web application is shown in Fig. 2, highlighting the structure and data flow. CQELedu provides a simple and user-friendly interface through Streamlit. The integration of LangChain and LLMs enhances efficiency and supports personalized learning experiences.

The user interface (UI) of CQELedu is shown in Fig. 3. The UI is designed to allow users to upload files and seamlessly generate questions. Supported file formats include PDF, DOCX, and TXT, which enhance user accessibility and convenience by accommodating diverse input types. Once a file is uploaded, it is processed using the PyPDF2 and docx2txt libraries to extract text, ensuring the efficient handling of different file formats.

The intuitive interface simplifies the process for users, making it easy to upload learning materials and generate customized questions based on their inputs. This design ensures a user-friendly experience and supports a broad range of educational requirements.

A GPT-4o mini LLM-based prompt chatbot is placed on the left sidebar to enable real-time interactions. The chatbot supports both question-and-answer and search functionalities throughout the system workflow. It enhances the learning experience by providing flexible interactions even after question generation and problem-solving. The chatbot interface on the sidebar can collapse or expand as required (Fig. 4), allowing adjustment options according to the user's convenience. This design optimizes screen space usage while maintaining ease of access for users.

After uploading a file, its name is displayed on the screen. Users can remove the file by clicking the "X" button next to its name, enabling them to upload a different file. Once the upload is complete, learners can choose the question type (multiple-choice with four options or short answer) to match their preferred learning style.

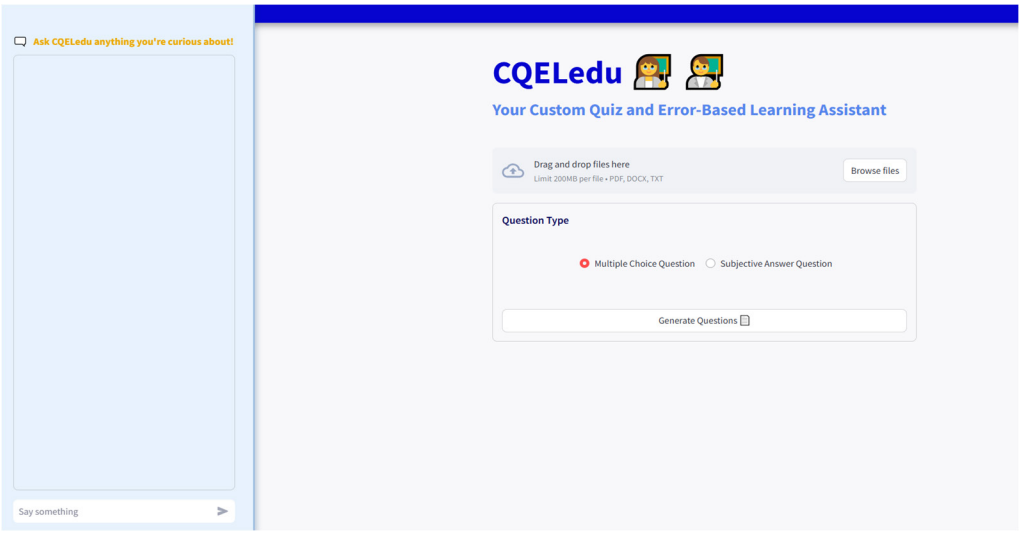


FIGURE 3. User interface of the web application.

Drag-and-drop functionality and radio buttons are incorporated to enhance usability and streamline the input process. After selecting the question type, users click the "Generate Question" button to initiate question generation. As shown in Fig. 5, once the button is clicked, a progress spinner and the status message "Generating questions…" appear on the screen, clearly indicating the ongoing process.

As shown in Figs. 6–7, the interface is simple and intuitive, enhancing user experience. Learners can easily understand their progress and current status, fostering an effective learning environment. A progress bar accompanies the generated personalized questions, allowing learners to intuitively monitor their progress during problem-solving. After entering their answer, learners can click the "Submit Answer" button to receive immediate feedback or the "Home" button to return to the initial screen.

When the "Submit Answer" button is clicked, the system instantly evaluates the answer and displays whether it is correct. If the answer is incorrect, the correct solution is provided, enabling learners to quickly understand and rectify their mistakes.

Clicking the "Home" button resets all uploaded files and question data, reloading the system to the initial screen.

Once all the questions are completed, the system displays a donut chart that visually represents the correct and incorrect answer rates, along with the final score on the screen.

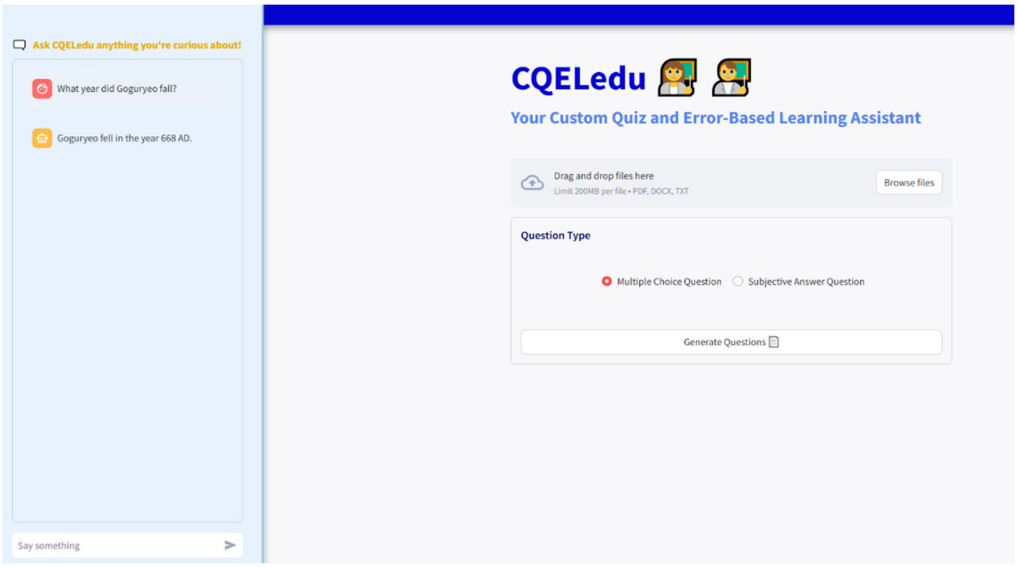
As shown in Fig. 8, clicking the "Generate Error-Based Questions" button at the top generates new error-based questions derived from keywords extracted from incorrect answers. While the topic shifts to focus on errors, the previously selected settings for the question format and number of questions remain unchanged. The subsequent steps for verifying correct answers and reviewing incorrect answers for error-based questions follow the same procedure as the original question-solving process. This design ensures a seamless and consistent learning experience, enabling learners to effectively address deficits in their understanding of the learning materials.

IV. EXPERIMENTS

The experimental design used to evaluate the performance of the CQELedu system is discussed in this section. The experiments were conducted using five PDF files covering topics related to Korean history, including economics, society, politics, policy, and culture. To ensure a comprehensive performance assessment, the evaluation included both MCQs and short-answer questions (SAQs). The performance of CQELedu was evaluated using three primary metrics: relevance, accuracy, and response time. These metrics were applied to assess various aspects of the system, including both basic and error-based question generation. In addition to the quantitative evaluation, a qualitative expert assessment was conducted by 8 Korean history experts to evaluate the relevance, error relevance, correctness, grammatical fluidity, and educational utility of the generated questions.

A. RELEVANCE

Relevance was evaluated by measuring semantic similarity between generated questions and the source material and by assessing how well the questions reflected the original content. Cosine similarity was measured between the generated MCQs and SAQs and their corresponding original documents. For this evaluation, 10 MCQs and 10 SAQs were generated per PDF file. This process was repeated 10 times, resulting in a total of 1,000 basic questions (500 MCQs and 500 SAQs). Similarly, 1,000 error-based questions (500 MCQs and 500 SAQs) were generated and evaluated. Similarity scores were measured separately for MCQs and SAQs per file. Thus, comparisons of relevance across PDF files and analysis of the consistency between different data types were



(a)



(b)

FIGURE 4. CQELedu interface with expandable and collapsible chatbot sidebar. (a) Sidebar expanded, showing the chatbot interface with real-time question and answer interaction. (b) Sidebar collapsed, hiding the chatbot interface.

performed. The analysis examined how the performance of the system varied depending on document content.

B. ACCURACY

Accuracy was evaluated by determining whether the error-based questions included core keywords related to the learners' incorrect answers.

For this evaluation, the number of incorrect answers was set to two, four, six, eight, and ten. Subsequently, questions were generated for each condition. Each condition was repeated twice, resulting in 500 MCQs and 500 SAQs.

The presence of core keywords was denoted using the following binary classification:

- 1 if a keyword was included.
- 0 if a keyword was not included.

This approach provided a quantitative measure of the reliability of error-based question generation. The variation in keyword inclusion rates with increasing numbers of incorrect answers was analyzed on a per-file basis to ensure consistency across different data types.

C. RESPONSE TIME

Response time was evaluated to measure the efficiency of the question-generation tasks. The time required to generate basic and error-based questions was recorded separately for MCQs and SAQs.

The performance of GPT-4o mini was compared with that of GPT-4o to highlight differences in time efficiency. Token costs were measured to assess both time and cost efficiencies. For error-based questions, generation requests were repeated 10 times under conditions with two, four, six, eight, and ten

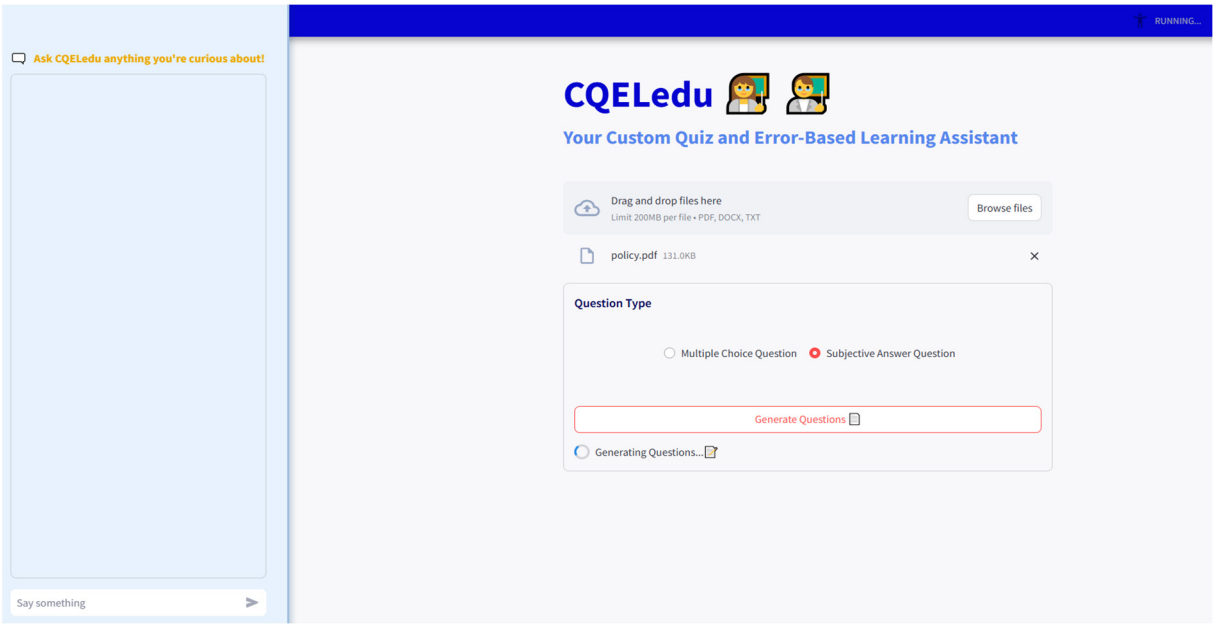


FIGURE 5. File upload and question generation.



FIGURE 6. Custom question generation and problem-solving.

incorrect answers, and the average response time was calculated for each condition. The evaluation examined whether the system maintained stability under increasing complexity and analyzed the differences in processing times across files with varying topics. This study provides insights into the robustness and adaptability of the system to diverse learning materials.

D. EXPERT EVALUATION

Expert evaluation was conducted to assess the content validity and educational utility of questions generated by the CQELedu system. With informed consent obtained in advance, the evaluation involved 8 professors specializing in education and Korean history, each with more than 20 years of academic experience. Each expert received 40 questions,

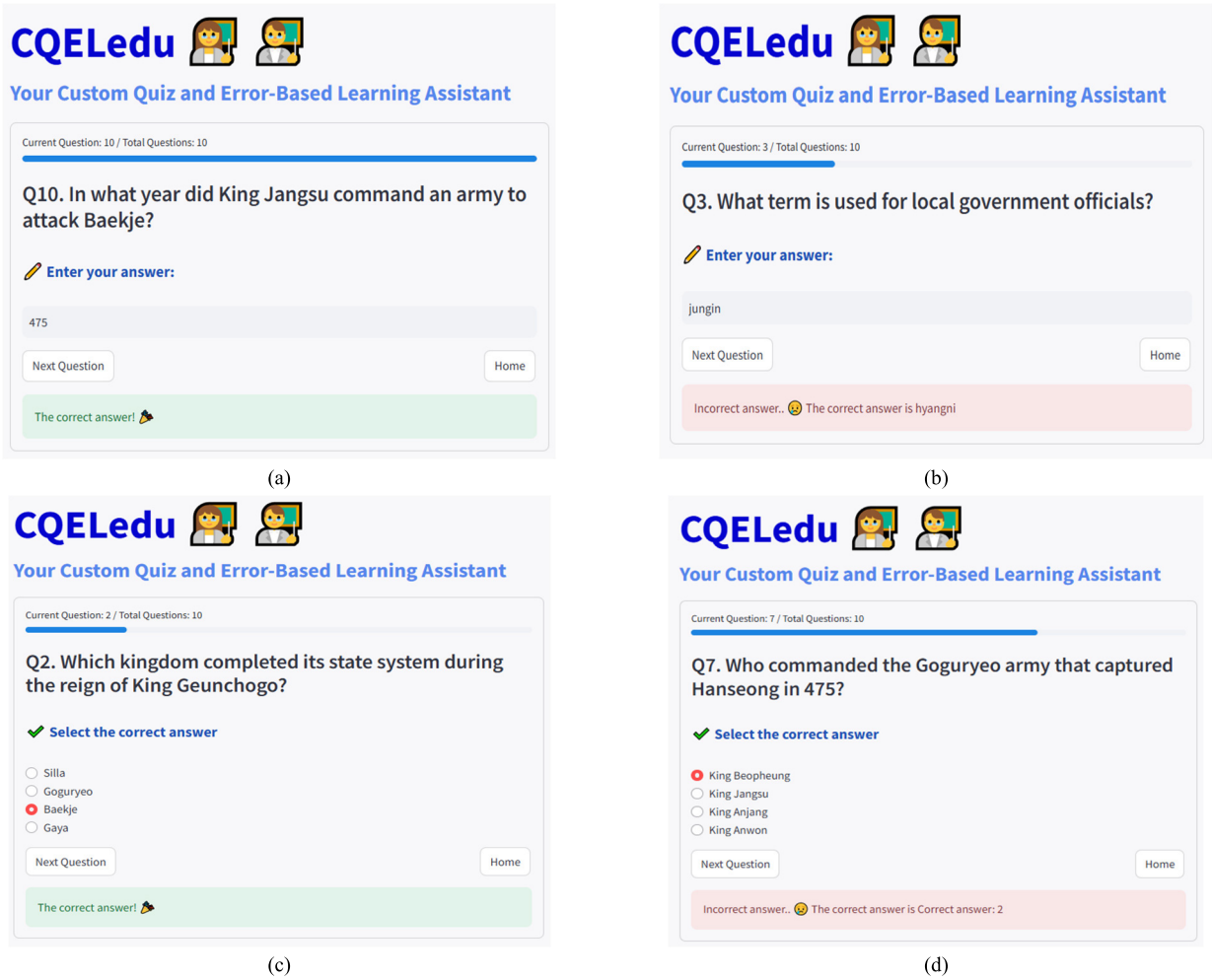


FIGURE 7. CQELedu problem-solving interface displaying various question types and feedback. (a) Short answer questions (SAQs) with a correct answer. (b) SAQs with an incorrect answer, providing the correct answer as feedback. (c) Multiple choice questions (MCQs) with a correct answer. (d) MCQs with an incorrect answer.

comprising 10 MCQs, 10 error-based MCQs, 10 SAQs, and 10 error-based SAQs.

The evaluation was carried out using a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The evaluation criteria were derived from previous studies on LLM-based question assessment and included the following:

- Relevance: Does the question adequately reflect the core concepts and content of the source material?
- Error Relevance: Does the question incorporate concepts meaningfully related to the learner's previous errors?
- Correctness: Is the question conceptually accurate and factually valid?
- Grammatical Fluency: Is the question expressed in a grammatically clear and natural manner?
- Educational Utility: Does the question effectively support learners in reviewing content, reinforcing concepts, or correcting misunderstandings?

The evaluation aligned its criteria with both question types and assessment objectives to validate the quality of the generated questions and to examine their educational contribution in error correction and conceptual reinforcement.

V. RESULTS

This section presents the performance evaluation results of the CQELedu system based on three key metrics: relevance, accuracy, and response time. The experiments were conducted on five topics: economics, politics, policy, culture, and society. Each metric of the system was comprehensively evaluated. Additionally, expert evaluation results are presented to provide a qualitative assessment of the pedagogical value and content quality of the generated questions.

A. RELEVANCE

Relevance was evaluated by measuring the semantic similarity between the questions generated by the CQELedu system and the original documents, using cosine similarity. The analysis was conducted separately for content and error-based questions categorized into MCQs and SAQs. The cosine similarity threshold was empirically adjusted based

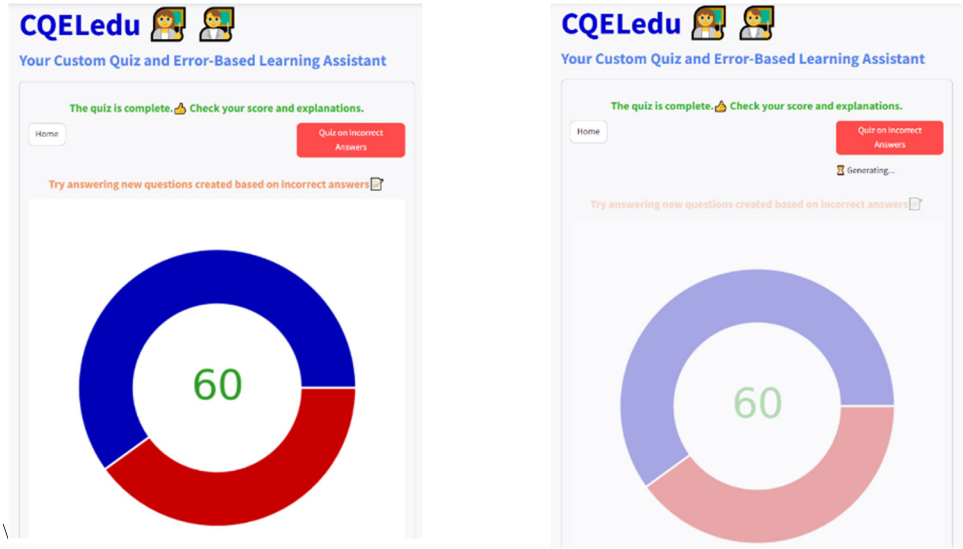


FIGURE 8. CQELedu result screen displaying quiz completion and error-based question generation. (a) Donut graph illustrating the correct (blue) and incorrect (red) answer distribution along with the final score. (b) State during error-based question generation, where questions are created based on incorrect answers, maintaining the same question type and count as the previous quiz.

on the average similarity score (0.86) calculated from the experimental results. The overall cosine similarity results for all the documents are presented in Fig. 9 and Table 2.

The results indicated high average cosine similarity scores ranging from 0.8 to 0.9 across both question types and for both content and error-based questions. High similarity scores across all categories consistently demonstrated the stable performance of the system. Notably, the similarity scores for error-based questions did not significantly decline, indicating that the system effectively preserved the core content of the documents, even when generating questions based on learners' incorrect answers.

These findings underscore the capability of the CQELedu system to maintain the integrity of source material while reflecting learners' error patterns in question generation.

B. ACCURACY

The experimental results showed that the accuracy of the error-based questions generated by the CQELedu system increased consistently with the number of incorrect answers. The experimental results are illustrated in Fig. 10, which provides a visual representation of the accuracy trends for the error-based questions.

For MCQs, accuracy rose from 0.85 when the number of incorrect answers was 2 to 0.99 when it reached 10. For SAQs, accuracy increased from 0.83 with 2 incorrect answers to 0.98 with 10 incorrect answers. In cases with a small number of incorrect answers, questions were sometimes generated outside the range of the core keywords, resulting in an accuracy of 0. However, as the number of incorrect answers increased, the generated questions increasingly incorporated relevant core keywords. In particular, when the number of incorrect answers was eight or more, the accuracy consistently exceeded 0.9, indicating a significant improvement.

Thus, while fewer incorrect answers could lead to broader question scopes and reduced accuracy, a higher number of incorrect answers allowed for a more precise integration of core keywords, improving accuracy. Additionally, the MCQs recorded higher accuracy across all conditions than the SAQs. Although both types of questions were generated based on the document, SAQs tended to include a broader range of keywords, leading to slightly lower accuracy. Thus, the CQELedu system was more effective in reflecting core keywords under specific conditions when generating MCQs.

In summary, the CQELedu system maintained consistently high accuracy in error-based question generation, regardless of the question type. As the number of incorrect answers increased, the reliability of the generated questions improved.

These findings highlight the ability of the system to effectively learn from learners' error patterns and accurately apply them to generate relevant questions.

C. RESPONSE TIME

The experiment compared the response times of OpenAI's GPT-4o and GPT-4o mini models for question-generation tasks in the CQELedu system. Both document and error-based question generation times were measured, and performance was evaluated based on average response times. The experimental results for the response times are shown in Fig. 11.

The analysis of response times for MCQs revealed that the GPT-4o mini model recorded an average response time of 8.1 s for document-based questions and 12.35 s for error-based questions. In comparison, the GPT-4o model took

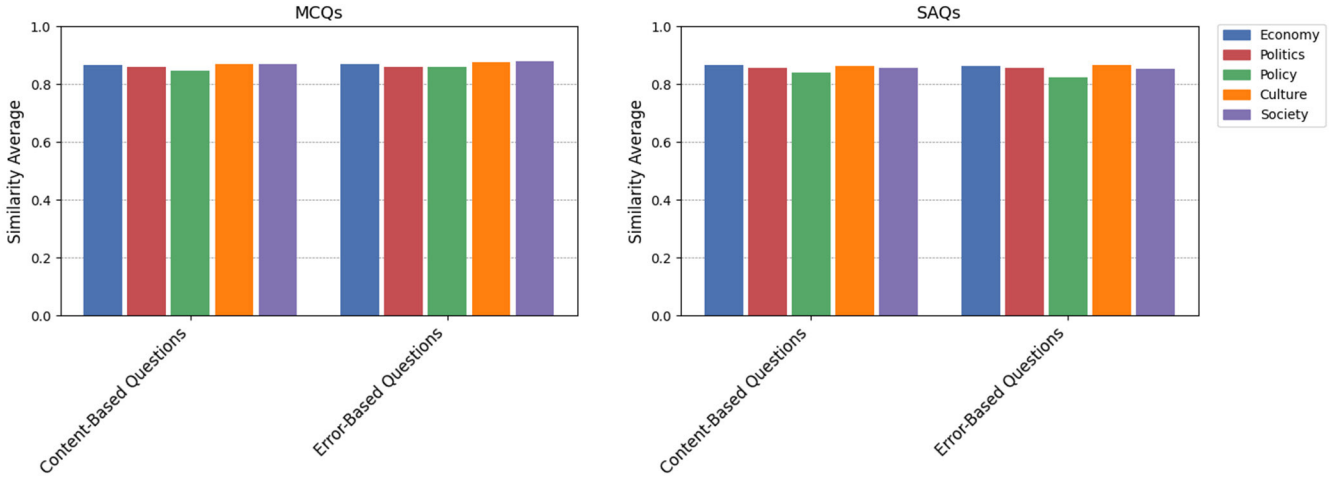


FIGURE 9. Comparison of question similarity averages for multiple choice questions (MCQs) and short answer questions (SAQs) across various topics (economy, politics, policy, culture, and society), categorized by content and error-based question types.

TABLE 2. Average similarity scores by category and question type.

Category	Average Similarity Score			
GPT-4o	MCQs		SAQs	
	Content-based	Error-based	Content-based	Error-based
Economy	0.8679	0.8691	0.8672	0.8630
Politics	0.8586	0.8604	0.8563	0.8582
Policy	0.8485	0.8608	0.8409	0.8240
Culture	0.8707	0.8762	0.8645	0.8652
Society	0.8708	0.8805	0.8577	0.8522

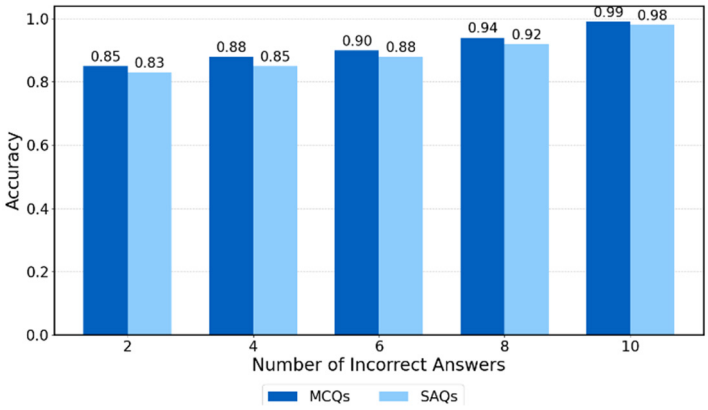


FIGURE 10. Accuracy comparison for error-based questions (MCQs vs. SAQs).

9.49 s for document-based questions and 13.99 s for error-based questions. These results indicate that the GPT-4o mini operates more efficiently and processes MCQs faster, owing to its lightweight and optimized design.

Similarly, for SAQs, the GPT-4o mini model outperformed the GPT-4o. For document-based questions, the GPT-4o mini recorded an average response time of 5.4 s, whereas the GPT-4o took 6.39 s. For error-based questions, GPT-4o mini recorded 6.3 s, compared to 8.89 s for GPT-4o.

These results suggest that GPT-4o mini is highly effective in generating concise question types, such as SAQs, where streamlined processing is critical.

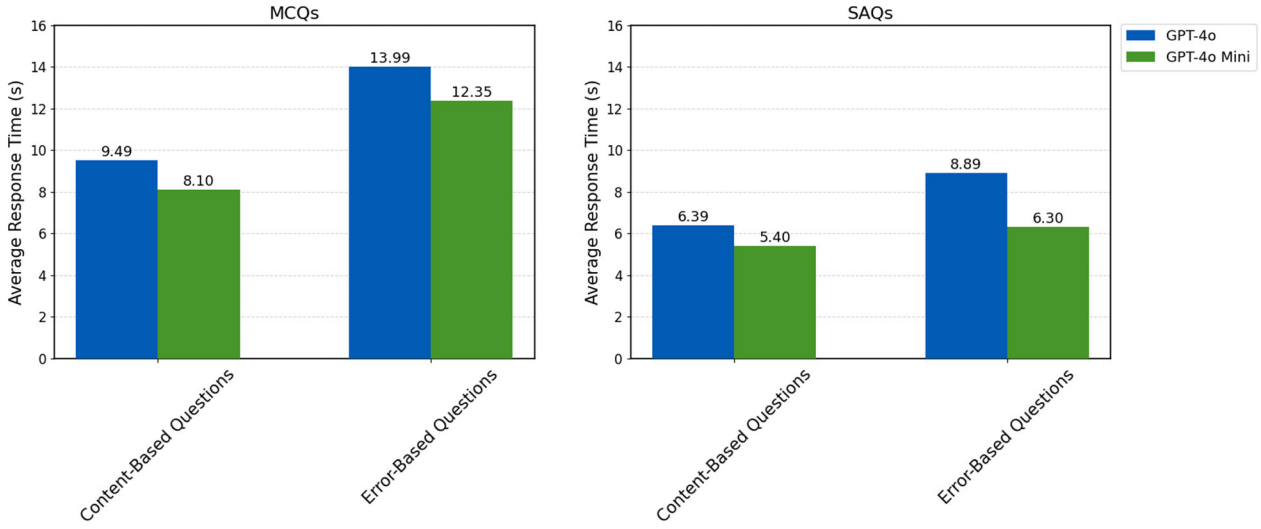


FIGURE 11. Average response times for content-based and error-based questions (MCQs and SAQs) Using GPT-4o and GPT-4o mini.

In summary, the comparison demonstrates that GPT-4o mini processes both document- and error-based questions more quickly than GPT-4o. This efficiency is particularly evident in shorter question formats, confirming the superiority of the GPT-4o mini for time-sensitive tasks.

The average response times for document and error-based questions generated by the CQELedu system using the GPT-4o mini model, with category-wise performance analyzed by question type are presented in Figs. 12.

The results demonstrated that, for both multiple-choice and short-answer questions, the response times were higher for error-based questions than for document-based questions. This is attributed to the additional computations required to incorporate the core keywords related to incorrect answers.

D. EXPERT EVALUATION

Figs. 13 and 14 illustrate the results of expert evaluations for MCQs and SAQs, respectively, based on five assessment criteria. Both MCQs and SAQs exhibited generally consistent evaluation patterns, with inter-rater agreement confirmed across four key criteria: relevance, conceptual validity, grammatical fluidity, and error relevance.

Inter-rater agreement was analyzed using Kendall's coefficient of concordance (W). Following the interpretation framework proposed by Landis and Koch, values of Kendall's W below 0.20 reflect slight agreement, values between 0.21 and 0.40 indicate fair agreement, values from 0.41 to 0.60 represent moderate agreement, and values above 0.60 denote substantial to almost perfect agreement among raters [45].

According to this interpretation scale, the analysis revealed statistically significant agreement among experts for all question types, with MCQs showing $W = 0.519$ ($p = 0.002$), SAQs showing $W = 0.443$ ($p = 0.007$), and all items combined showing $W = 0.469$ ($p < 0.001$), indicating moderate agreement among raters.

In both formats, the CQELedu system demonstrated stable quality by effectively incorporating core concepts and learner error data into the question generation process. Notably, relevance received the highest overall rating among all evaluated criteria, highlighting the system's strength in aligning generated content with the key instructional concepts.

Error relevance was relatively lower, particularly for MCQs, likely because the multiple options broadened the question scope and reduced the direct semantic connection between error keywords and generated questions. This result indicates that the semantic link between error contexts and instructional content was not always adequately preserved, highlighting the need to improve the conceptual alignment and contextual design of error-based questions.

Grammatical fluidity received a lower evaluation, suggesting that some AI-generated questions were perceived by experts as unnatural or grammatically incomplete. These findings indicate the need to improve linguistic naturalness and grammatical completeness in AI-based question generation.

Educational utility was rated higher for MCQs than for SAQs, reflecting format-specific differences in feedback and error reflection. This finding emphasizes the need to optimize system design based on the characteristics of each response format.

VI. DISCUSSION

This study demonstrated that the CQELedu system is effective as a personalized learning support tool, showing high similarity and answer accuracy in both document-based and error-driven question generation. The generated questions reflected the context of the original documents well, and stable performance was maintained regardless of question type or topic. Compared to previous approaches, this system is distinguished by moving beyond simple question banks or

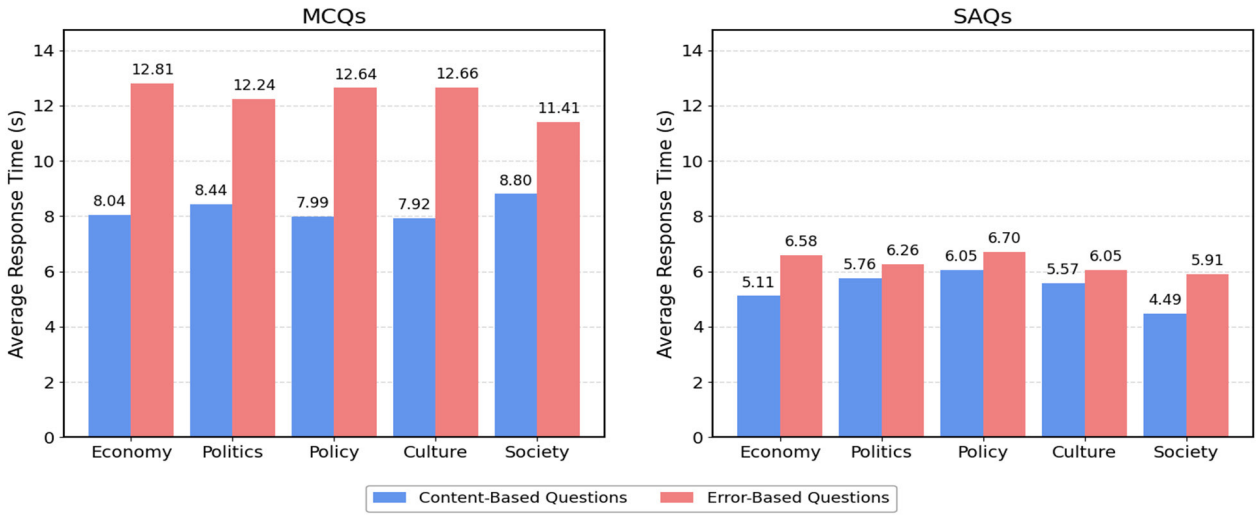


FIGURE 12. Average response times for content- and error-based questions (MCQs and SAQs) across topics generated by the CQELedu system.

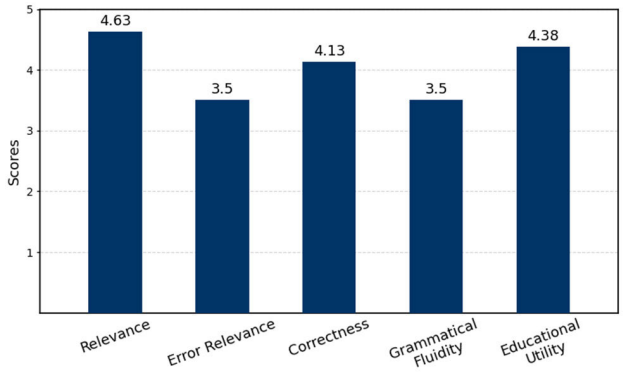


FIGURE 13. Average evaluation scores for MCQs across five criteria by expert evaluators.

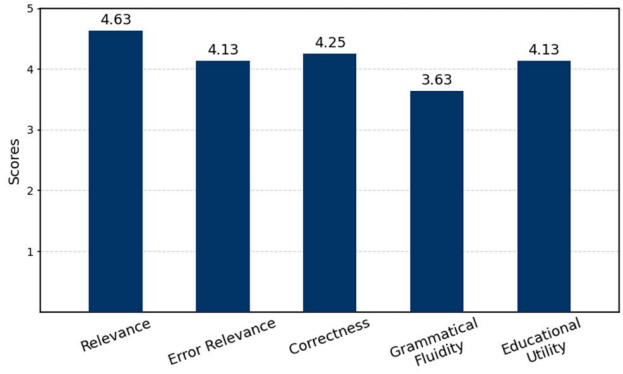


FIGURE 14. Average evaluation scores for SAQs across five criteria by expert evaluators.

one-time question generation, instead providing an adaptive learning environment through error-driven iterative practice. The combination of LangChain and GPT-4o mini enabled fast response times and low operational costs, while supporting real-time interaction for improved practicality.

Despite these meaningful achievements, several limitations remain. First, the system was designed primarily for English, and question generation and evaluation experiments were also conducted mainly on English texts. This results in linguistic constraints and limits the generalizability and applicability of the system to various language environments and actual educational settings. Future work should expand multilingual support to improve linguistic scalability and accessibility for diverse learners.

In addition, the learning materials used in the experiments were limited to topics related to Korean history, and thus the scalability and generalizability to other domains such as science, mathematics, or healthcare have not been fully verified. Additional experiments using educational content from various fields are needed to further validate the system's versatility and robustness.

From a technical perspective, while the system used fixed-format LangChain prompt templates, LLMs are subject to prompt sensitivity, meaning output may vary with subtle differences in wording, even when the meaning is the same. This study did not conduct an immediate robustness evaluation regarding prompt sensitivity, and the stochastic nature of LLM output remains an important technical limitation. In particular, it may be difficult to ensure full consistency and stability in actual educational environments.

The system also adopted a Retrieval-Augmented Generation (RAG) structure to structurally mitigate hallucination issues. However, given the generative characteristics of large language models, it is difficult to completely eliminate potential hallucination.

These limitations are particularly relevant to the reliability of formative assessments generated by the system.

Inconsistent outputs resulting from prompt sensitivity may unintentionally alter the difficulty or content validity of questions, thereby affecting the stability of evaluation results.

Moreover, hallucinations can compromise the validity of formative assessments by introducing information that is not supported by the original learning materials.

To enhance reliability, future versions of CQELedu will incorporate prompt-stability verification, hallucination-filtering mechanisms that cross-check generated content with retrieved evidence, and difficulty calibration to maintain consistent cognitive complexity across sessions.

Through these improvements, the system aims to further reinforce the reliability and fairness of AI-based formative assessments.

Finally, although CQELedu was designed for practical deployment in real educational settings, the system still carries inherent ethical risks common to AI systems, such as the generation of inappropriate content, bias, and personal information leakage.

To address these issues, harmful content filtering prompts were applied, and user input of personal information was minimized; however, such technical measures alone cannot completely resolve all risks. To further address the issue of bias, it is necessary to collect and apply data from a wide range of languages, topics, and domains, improve prompt design, and strengthen the process of bias diagnosis.

Future research should also involve more sophisticated privacy protection mechanisms and the development of ethical usage guidelines. In particular, to enhance the practical utility of CQELedu, we plan to develop multilingual support. We also aim to further strengthen adaptive learning by dynamically analyzing learner responses and performance to adjust question difficulty and feedback.

Moreover, because the contents of documents used for question generation are transmitted to external servers, there is a potential risk of personal information leakage. Future research will aim to enhance privacy protection and system independence by improving the system to operate in a local environment using open-source LLM models. Through such advancements, CQELedu can evolve into a highly reliable and classroom-friendly intelligent learning system.

VII. CONCLUSION

This study introduces CQELedu, a web application designed to generate questions based on learning materials and provide error-driven questions for facilitating personalized learning. CQELedu uses GPT-4o mini and LangChain to generate SAQs and MCQs while analyzing learners' incorrect answers to create a customized learning environment. Additionally, the application employs Streamlit to deliver a user-friendly interface, enhancing accessibility and convenience for learners.

System performance was evaluated through three key metrics: relevance, accuracy, and response time. Both document and error-based questions demonstrated high semantic consistency with the original learning materials, achieving average cosine similarity scores between 0.8 and 0.9. Further, as the number of incorrect answers increased, the system generated questions with improved accuracy; MCQs achieved higher accuracy than SAQs. However, when the number of incorrect answers was low, the system occasionally produced questions with broader keyword coverage. Response time analysis revealed that GPT-4o mini consistently outperformed GPT-4o, showcasing the efficiency of the lightweight model in completing question-generation tasks more quickly.

In conclusion, CQELedu effectively generates personalized questions by reflecting learners' error patterns through error-driven question generation. The system's effectiveness, validated through quantitative and qualitative assessments, highlights its potential to support dynamic learning environments in standardized testing contexts, such as certification exams and school assessments. This study underscores the possibilities of AI-based educational technology in fostering more adaptive and personalized learning experiences.

FUTURE WORK

Future research should investigate the scalability of CQELedu and its applicability across various learning domains, aiming to develop a more sophisticated and comprehensive learning support system. In particular, the system will be further enhanced to support domain-specific question generation by incorporating educational content from various fields such as science, mathematics, and healthcare. Through this enhancement, the system can be improved to ensure that it is not limited to a specific field, but can be utilized in a wide range of educational contexts. Furthermore, a multilingual learning environment will be implemented by supporting major languages in addition to English, or by allowing users to select their preferred language. By integrating learning materials from diverse fields and supporting multiple languages, the system's universality and scalability could be enhanced, facilitating its application across a wider range of educational environments beyond specific subject areas.

Additionally, the system could be improved by expanding user options, enabling learners to freely select the type and number of questions. This enhancement would increase the flexibility of personalized question generation, enabling the system to deliver questions optimized for individual learning goals and proficiency levels, thereby maximizing learning efficiency.

Furthermore, in error-driven question generation, future improvements will go beyond simple keyword-matching approaches by incorporating advanced NLP techniques such as concept extraction and dependency parsing. These deep NLP methods can enable more precise analysis of errors and facilitate the generation of supplementary questions that are more closely related to the underlying concepts.

Feedback functionality could be strengthened to continuously analyze learners' understanding levels and learning patterns, ultimately proposing customized learning paths. In particular, the system could introduce features to deeply analyze repeated error patterns, automatically recommend

supplementary learning materials, and generate additional questions. By integrating adaptive learning into the system, these features would dynamically adjust learning content and pathways to each learner's needs, helping them effectively address their weaknesses while maintaining motivation for continuous improvement.

Furthermore, considering the nature of LLM-based systems, future research should also address the issues of prompt sensitivity and potential hallucinations. While CQELedu has demonstrated stable performance under various conditions, this study did not fully explore robustness against variations in prompt phrasing.

Another important direction for future development involves regulating the difficulty level of generated questions, particularly for short-answer (subjective) items.

In the current version, question complexity was not explicitly calibrated, and some generated items tended to be overly specific or factual in nature.

To enhance educational validity and ensure balanced assessments, future iterations of CQELedu will incorporate automatic difficulty calibration based on linguistic complexity, conceptual depth, and cognitive demand.

This enhancement will enable the system to generate and label questions across multiple difficulty tiers (easy, moderate, and challenging), thereby supporting more accurate formative evaluation and promoting adaptive learning.

Since prompt formulation can significantly influence LLM outputs, future system improvements will include experiments that evaluate prompt stability and response consistency. In addition, mechanisms for detecting and mitigating hallucinations will be integrated to enhance the reliability and educational validity of the generated questions and feedback. By conducting comprehensive robustness checks and implementing hallucination detection strategies, the system's credibility and trustworthiness in educational contexts can be further strengthened.

Through these advancements, CQELedu is expected to evolve into a more precise and effective personalized learning support tool, contributing significantly to the future development of intelligent education systems. By offering a personalized and dynamic learning environment optimized for standardized contexts, such as certification exams and school assessments, this study opens new avenues for enhancing educational technology and improving learning outcomes. As a tool that sets a new direction for personalized learning, CQELedu is positioned to play a meaningful role in advancing next-generation educational systems.

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